

KinaseCanvas: An Integrated Resource for Curated and Novel Kinase-Substrate Interactions

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1. Abstract

Protein phosphorylation is a cornerstone of human physiology, regulating pathways critical for development, metabolism, and disease. While curated databases like PhosphoSitePlus and iPTMnet provide high-confidence kinase-substrate interactions (KSIs), they remain biased toward well-studied signaling nodes, leaving the vast majority of the “dark” kinome unconnected. Recent specificity atlases have begun to fill this gap by profiling motif specificities for nearly all human kinases, yet these raw predictions are often too dense and noisy for direct biological interpretation. Here, we present KinaseCanvas, an integrated resource that places these motif predictions within the context of high-confidence curated data. By applying stringent biological filters for subcellular co-localization and phosphosite functional importance, we distill tens of millions of raw predictions into 21,916 high-confidence novel interactions and integrate them with 8,364 curated interactions, yielding 30,280 distinct kinase-substrate pairs. This integration increases the share of interactions involving understudied kinases roughly 4.6-fold, from 3.4% to 15.7%, and provides a context-aware framework for hypothesis generation, illuminating poorly characterized regions of the human kinome.

Database URL: <https://cujoisa.github.io/KinaseCanvas/>

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24 2. Introduction

25 Protein phosphorylation is a pervasive regulatory mechanism that controls
26 virtually every major cellular process. Public repositories now catalogue well
27 over 200,000 human phosphosites distributed across the large majority of the
28 human proteome [1, 2], and modelling that accounts for the incomplete and
29 biased coverage of tryptic phosphoproteomics places the true totals higher
30 still [2, 3]. Estimates of how many of these sites are individually functional
31 are considerably lower [4, 5, 6], so the operative problem is not enumeration
32 but prioritization. Mapping which phosphorylation enzyme (kinase) modifies
33 which protein (substrate) is therefore essential for understanding signal trans-
34 duction, drug response and disease mechanisms, but this map remains far from
35 complete. Over the past two decades, substantial effort has been put into
36 building reference resources for kinase-substrate interactions (KSIs). Databases
37 such as PhosphoSitePlus, EPSD and iPTMnet curate experimentally observed
38 phosphosites and, where possible, annotate the upstream kinases based on low-
39 throughput biochemistry, targeted phosphoproteomics and manual literature
40 curation [1, 7, 8]. Additional resources including Phospho.ELM, PhosphoNet-
41 works and SIGNOR provide complementary KSI annotations derived from both
42 focused and high-throughput studies [9, 10, 11], although these differ substan-
43 tially in scope and currency: Phospho.ELM has not been updated since 2010,
44 and PhosphoNetworks reports kinase-substrate relationships measured on pro-
45 tein microarrays without resolving the modified residue [12, 10]. PhosphoNET
46 additionally provides sequence-based predictions of kinase-phosphosite pairings
47 for 486 human protein kinases, using scoring matrices that span the -7 to +7
48 positions around the phosphoacceptor, alongside curated sites [13]; the com-
49 panion KiNECTOR resource (<http://www.kinector.ca>) aggregates over 25,000
50 human kinase-substrate-phosphosite relationships into navigable pathway maps.
51 However, these curated datasets are heavily skewed towards a small subset of
52 well-studied kinases, substrates and pathways, leaving large regions of the human
53 kinome (including understudied “dark” kinases) sparsely connected or completely
54 unmapped [14].

55 In parallel, unbiased experimental and computational approaches have recently
56 produced near-comprehensive “specificity atlases” for the human kinome. Using
57 peptide-library based specificity profiling, Johnson et al. systematically deter-
58 mined optimal substrate motifs for most human serine/threonine kinases and
59 used these motifs to score reported human phosphosites [15]. More recently,
60 Yaron-Barir et al. generated an analogous atlas for the human tyrosine kinome
61 [16]. Together, these resources greatly expand the theoretical kinase-phosphosite
62 search space: for each phosphosite, nearly all known kinases receive a motif-based
63 compatibility score, yielding tens of millions of candidate KSIs. While these
64 atlases have already been shown to recover known signaling relationships and
65 highlight kinase dysregulation in disease contexts [15, 16], their full potential as a
66 systems-level KSI prior remains underexploited. Current visualization tools, such
67 as the Kinase Library [15] and earlier motif scanners such as Scansite [17], focus
68 on site-level scoring and lack interactive network visualization of kinase-substrate

relationships. Furthermore, while methods like Kinase-Substrate Enrichment Analysis (KSEA) are widely used to infer kinase activity, they typically rely on curated databases or require manual integration of specificity matrices.

Moreover, profiling thousands of phosphosites against hundreds of kinases generates an enormous combinatorial prediction space: most sites have multiple plausible motif matches, and naïve motif scanning across the phosphoproteome leads to high false-positive rates and spurious kinase-substrate couplings [18, 19]. A major challenge is therefore that motif-based scores alone do not ensure that a kinase-substrate pair is physically or functionally plausible. Kinases and substrate pairs must co-localize in the cell, be expressed in the same tissues or conditions, and the modified site itself must be known or predicted to have some functional consequence. To address this, recent studies have demonstrated the value of integrating orthogonal data layers. In particular, Ochoa et al. developed a machine-learning approach that aggregates features ranging from evolutionary conservation and structural constraints to regulation across thousands of phosphoproteomics experiments. Their functional score effectively discriminates regulatory sites from non-functional ‘noise,’ substantially improving the biological interpretability of phosphoproteomic datasets [5, 20]. When combined with orthogonal constraints such as subcellular localization, these functional indicators could provide a powerful filter for identifying physiologically relevant kinase-substrate relationships.

Here, we build on these advances to construct an integrated resource of human kinase-substrate relationships that combines literature-curated KSIs with motif-based predictions from the serine/threonine and tyrosine kinome specificity atlases. We apply a set of simple but stringent biological filters based on (i) motif rank, (ii) subcellular co-localization using the Human Protein Atlas (HPA), and (iii) a data-driven phosphosite functional score [5, 21, 15, 22]. An overview of the data integration and filtering strategy is shown in Figure 1(A), including an example TRAF2 and NCK-interacting protein kinase (TNIK)-centred subnetwork. The resulting network is accessible through the KinaseCanvas Explorer, an interactive web application that allows users to query specific kinases or substrates, visualize their local interaction neighbourhoods, and seamlessly filter connections based on curation status or functional evidence. This unified framework enables researchers to distinguish novel from previously reported interactions and to systematically assess coverage of both well-studied and dark kinases.

Alt text: Workflow diagram showing integration of curated Kinase-Substrate sets with motif predictions, co-localization filters, and functional scores.

3. Materials and Methods

3.1. Curated kinase-substrate interactions

We first assembled a non-redundant reference set of human kinase-substrate-site relationships (“curated KSIs”) from multiple databases. Primary sources

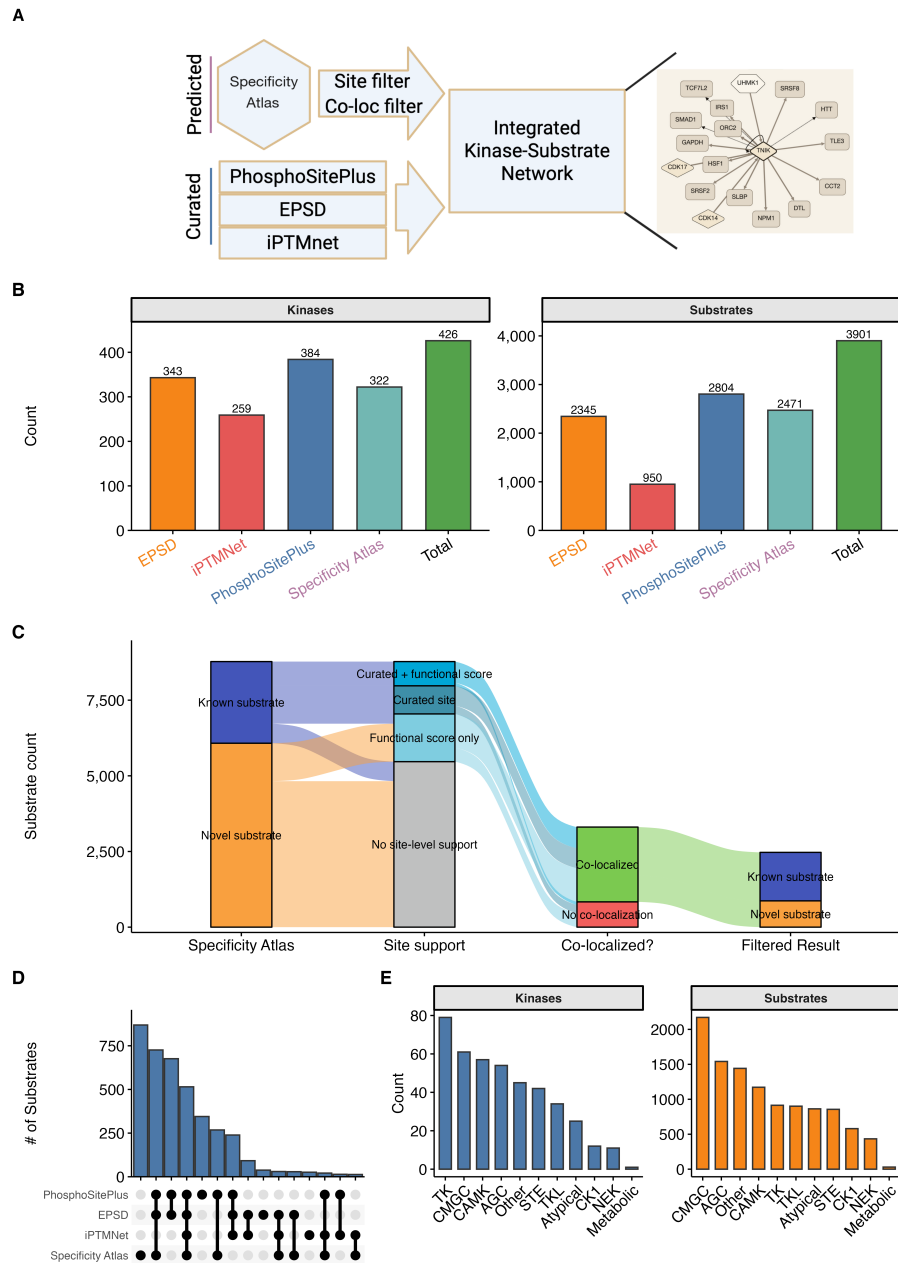


Figure 1: Construction of the integrated kinase-substrate network. A) Overview of the integration strategy. Motif-based predictions from the specificity atlas are filtered by site-level and co-localization criteria and merged with curated KSIs from PhosphoSitePlus, EPSP and iPTMnet to generate an integrated kinase-substrate network. An example TNK1-centred neighbourhood from the KinaseCanvas Explorer illustrates how curated (dark) and predicted (light) interactions are combined. B) Numbers of kinases (left) and substrates (right) contributed by each source and in the integrated total. C) Alluvial diagram tracking substrates through successive filtering stages: known vs. novel status, site-level support, and co-localization, to the final high-confidence set. The diagram is conditioned on substrates already retained by the initial motif and promiscuity filters and is not a sample of the human phosphoproteome. D) UpSet plot showing overlap of substrates across the four main KSI sources, highlighting both unique and shared substrate sets. E) Distribution of kinases (left) and substrates (right) across major kinase families in the integrated network.

111 included PhosphoSitePlus (PSP), the Eukaryotic Phosphorylation Site Database
112 (EPSD) and iPTMnet, which catalogue experimentally observed phosphosites
113 and often annotate upstream kinases based on low-throughput experiments or
114 kinase assays [23, 7, 8].

115 As a starting point, we downloaded the pre-assembled curated KSI table from
116 the KiNet web portal, which aggregates KSIs from PhosphoSitePlus, EPSD and
117 iPTMnet and provides kinase annotations [24]. We then re-processed this table to
118 harmonize identifiers and used available site-level functional annotations needed
119 for our analyses. Briefly, we extracted the kinase gene symbol, substrate protein,
120 phosphosite (modified residue and position), source database and/or PMID.
121 Gene and protein identifiers were harmonized by mapping to official HGNC gene
122 symbols and UniProt accessions (current versions as of January 10, 2026). When
123 multiple isoforms existed, phosphosites were mapped to the reviewed (Swiss-Prot)
124 sequence based on the human proteome when possible. Phosphosite notation
125 was standardized to a single-letter residue code plus position (e.g. “S123”).
126 Redundant entries arising from overlap across databases were collapsed, while
127 preserving provenance information (all contributing databases and PMIDs).

128 Across PhosphoSitePlus, EPSD, iPTMnet and the specificity atlas we ultimately
129 obtained 13,448 unique phosphosites, with PhosphoSitePlus contributing the
130 largest number (9,339 sites) and EPSD (7,525 sites) and the specificity atlas
131 (5,308 sites) contributing substantial additional coverage (Supp. Figure 1(A)).
132 The overlap structure of these site sets, including sites unique to individual
133 resources and those shared across multiple databases, is summarized in Supp.
134 Figure 1(B). The resulting curated KSI set serves as a reference baseline for all
135 subsequent analyses, and coverage of kinases and substrates contributed by each
136 source is shown in Figure 1(B).

137 3.2. Kinome-wide motif-based substrate predictions

138 To substantially expand beyond known KSIs, we integrated motif-based pre-
139 dictions from the human kinome specificity atlases for serine/threonine and
140 tyrosine kinases. Johnson et al. used position-specific scoring matrices derived
141 from peptide library screens to define optimal substrate motifs for the vast
142 majority of human serine/threonine kinases and subsequently scored reported
143 human phosphosites against each kinase-specific motif [15]. These matrix-based
144 approaches have been complemented by more recent deep-learning architectures
145 for phosphorylation site prediction [25]. More recently, Yaron-Barir et al. gen-
146 erated an analogous atlas for the human tyrosine kinome [16]. We and others
147 collectively refer to these resources as the “specificity atlas”.

148 We obtained the per-kinase, per-phosphosite prediction scores from the sup-
149plementary materials. For each phosphosite (UniProt accession + position),
150 the atlases provide a quantitative score or percentile rank reflecting how well
151 the local sequence matches the optimal motif for each kinase [15, 16]. The
152 complete, unfiltered specificity dataset—containing motif ranks, percentiles, and
153 promiscuity indices for nearly 28 million kinase-substrate-site combinations—is

provided as a supplemental resource (Supp. Table 1). From this prediction matrix, we extracted high-confidence motif-based KSIs using a percentile threshold. Following the original studies, we focused on phosphosite-kinase pairs in which the site ranked within the top 10% of all phosphosites scored for that kinase (≥ 90 th percentile) [15, 16]. This conservative cutoff retains sites that are strongly preferred by the kinase’s motif and reduces the inclusion of weak, low-specificity matches. Because a single stringent tier structurally excludes kinases whose specificity is dominated by docking interactions rather than by the catalytic cleft (see Discussion), we additionally release a relaxed tier comprising kinase-site pairs between the 75th and 90th motif percentiles that satisfy the same contextual criteria. This layer is released as a separate, labelled table (see Data Availability) and is not part of the high-confidence set.

To further mitigate the problem of phosphosites that match the motifs of many kinases, we utilized the “promiscuity index” provided for each scored phosphosite, defined as the number of kinases for which that site ranks at or above the 90th percentile. Sites at or above the median promiscuity index (21 kinases) were excluded from the motif-based prediction set, as a site compatible with many kinases’ motifs carries little information about which of them acts on it. The remaining set of motif-supported KSIs was then passed to contextual filters described below.

3.3. Contextual filtering

3.3.1. Subcellular co-localization

Since kinase activity is constrained by spatiotemporal co-localization with substrates, we filtered motif-based KSIs based on subcellular localization. We used the HPA subcellular annotations, which provide antibody- and imaging-based localization calls for thousands of human proteins across 30 subcellular structures [22] (HPA subcellular location dataset, downloaded 10 January 2026). The mapping of gene-level subcellular compartmentalization used in this study is provided as a supplemental table (Supp. Table 2). For each kinase and substrate, we retrieved all reported compartments annotated at the “Enhanced”, “Supported” or “Approved” reliability levels, excluding “Uncertain” calls. For every predicted kinase-substrate pair, we then computed the intersection of kinase and substrate localization sets and recorded the number of shared compartments, which is released as a continuous column rather than only as a binary flag. Pairs with at least one shared compartment were considered co-localized and retained; pairs with no compartmental overlap were flagged as unlikely to occur in vivo and removed from the high-confidence set. Proteins absent from HPA necessarily have no shared compartment and are therefore treated as non-co-localized, so missing annotation and genuine spatial separation are not distinguished by this filter. The distribution of the fraction of motif-predicted substrates that co-localize with each kinase is shown in Figure 2(A); the median kinase retains 37.2% of its predicted substrates after applying this co-localization filter. We note that HPA localizations are neither exhaustive nor condition-specific, so some genuine context-dependent interactions may be lost. However, requiring at

198 least one documented shared localization substantially increases the plausibility
199 of retained KSIs.

200 3.3.2. *Phosphosite functional score*

201 To prioritize phosphosites with higher likelihood of regulatory impact, we
202 integrated the phosphosite functional score developed by Ochoa et al., who
203 re-analysed thousands of phosphoproteomics experiments to build a reference
204 human phosphoproteome and then trained a gradient boosting model on sites of
205 known function to integrate 59 features, grouped as mass-spectrometry evidence,
206 phosphosite regulation, structural environment and evolutionary conservation,
207 into a single score of functional relevance [5]. The resulting score ranges from 0
208 to 1, with higher values indicating a greater probability that a site is functionally
209 important.

210 We downloaded the pre-computed human phosphosite functional scores and
211 mapped them to our phosphosite identifiers (provided as Supp. Table 3). For
212 each predicted KSI, we annotated the substrate site with its functional score
213 and designated a site as “likely functional” if its score was ≥ 0.5 . This is the
214 threshold recommended by the authors of the score, who report that at a 0.5
215 cut-off approximately 10% of the phosphoproteome is retained while 50% of
216 true-positive functional sites are recovered, making it a useful cut-off for initial
217 prioritization [5]; comparable thresholds have been applied in subsequent work
218 [16]. Because any fixed threshold is arbitrary at its boundary, we additionally
219 report a sensitivity analysis across the range 0.3-0.7 and release a continuous
220 confidence score (see below). As expected, tyrosine sites were markedly enriched
221 for site-level support compared to serine or threonine sites (36.1% vs. 12.8% and
222 15.0% of unique sites, respectively; Figure 2(B)). Here, as in Figure 2(B), a site
223 counts as supported if it is either already curated or carries a functional score
224 above the cutoff; restricting to functional-score support alone among sites that
225 carry a score gives 26.5%, 7.7% and 8.6%. The difference between residue classes
226 is highly significant ($\chi^2 = 1848.1$, $df = 2$, $p < 2 \times 10^{-16}$; $n = 5,132$ tyrosine,
227 34,754 serine and 7,844 threonine sites). When stratified by source database,
228 the proportion of scored sites classified as likely functional ranges from 45.4%
229 (specificity atlas layer) and 47.4% (PhosphoSitePlus) through 50.1% (EPSPD) to
230 62.2% (iPTMnet) (Figure 2(C) and Supp. Figure 1(C)). The atlas-derived layer
231 is not enriched relative to the curated sources because, under the disjunctive
232 site-level criterion described below, a site can enter it on the strength of curation
233 alone; the comparison is therefore most informative for the curated sources,
234 which are not filtered on this score. Unless otherwise noted, analyses focusing
235 on functional sites consider only motif-based KSIs whose substrate sites passed
236 this functional score threshold.

237 Alt text: Statistical distribution of phosphosite functional scores across different
238 residue types (Y, S, T) and source databases.(A) shows co-localization fraction
239 by kinase;(B) shows functional score enrichment for tyrosine sites.

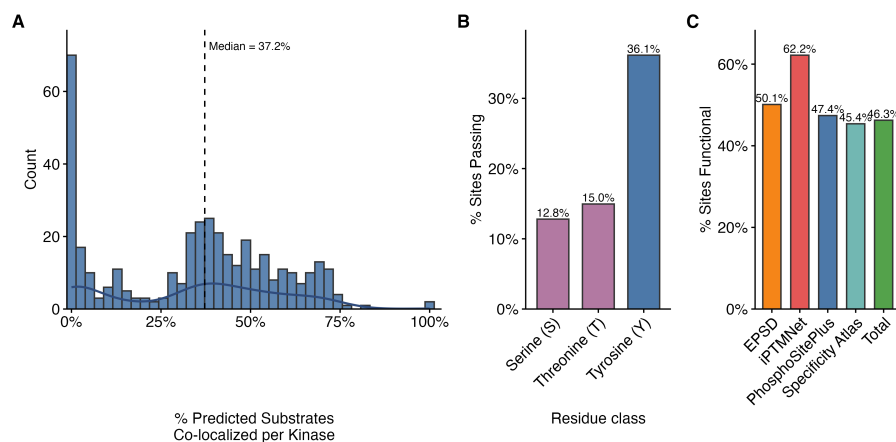


Figure 2: **Specificity Atlas Filtering Results.** A) Distribution of subcellular co-localization percentages for predicted kinase-substrate pairs per kinase. Co-localization was determined using HPA subcellular location data, requiring at least one shared compartment between the kinase and substrate; percentages are computed over each kinase’s predicted substrates, counting those without HPA annotation as non-co-localized, as the filter does. The dashed line indicates the median co-localization (37.2%). B) Percentage of predicted sites passing filtering criteria across residue classes (Serine, Threonine, and Tyrosine). “Pass” is defined as a site being either previously curated in primary databases (PhosphoSitePlus, EPSPD, iPTMNet) or possessing a high functional score (Ochoa score ≥ 0.5). C) Percentage of scored sites classified as likely functional (functional score ≥ 0.5) in each data source.

240 *3.4. Integration of motif, localization and function*

241 Combining the motif, promiscuity, localization and functional filters, we defined
242 a high-confidence predicted KSI set as those kinase-substrate-site triplets that
243 satisfy all of:

- 244 1. Motif support: the substrate site ranks in the top 10% of all phosphosites
245 scored for that kinase (≥ 90 th percentile) in the relevant serine/threonine
246 or tyrosine atlas.
- 247 2. Specificity: the site falls below the median promiscuity index, excluding
248 sites that are top-ranked by many kinases.
- 249 3. Co-localization: kinase and substrate share at least one HPA-annotated
250 subcellular compartment.
- 251 4. Site-level support: the substrate site either has a phosphosite functional
252 score ≥ 0.5 or is already curated in PhosphoSitePlus, EPSD or iPTMnet.

253 Criterion 4 is deliberately a disjunction rather than a conjunction. A site that
254 has already been curated from the primary literature carries direct experimental
255 evidence of relevance, so requiring it to additionally clear a machine-learned
256 functional-score threshold would discard well-established sites on the basis of
257 a predictor. Curated and functional-score support are reported as separate
258 columns so that either can be required independently.

259 The effect of these filters on the substrate set is visualized as an alluvial
260 diagram in Figure 1(C), with per-stage substrate counts given in Supp. Table 4.
261 It tracks substrates from known vs. novel status, through site-level support and co-
262 localization, to the final set. Although some true interactions will undoubtedly
263 be excluded (e.g. transient co-localization not captured by HPA, functional
264 sites below the threshold, or kinases with context-dependent specificity), we
265 deliberately favour specificity over sensitivity to generate a conservative, high-
266 confidence prediction set. Each predicted KSI is also cross-referenced to the
267 curated KSI list; interactions present in curated databases are flagged as “known”,
268 whereas motif-only interactions with no curated evidence are classified as novel
269 predictions contributed by this resource.

270 *3.5. Species and isoform handling*

271 The resource is restricted to human proteins (NCBI taxonomy 9606). Every
272 protein in the curated KiNet table is a reviewed (Swiss-Prot) human UniProt
273 entry, and we performed no cross-species orthologue mapping of our own, so a
274 site enters the resource only if a contributing database reports it on the human
275 protein. All phosphosite positions refer to the canonical UniProt sequence of the
276 substrate: neither the curated table nor the specificity atlases use isoform-specific
277 accessions. Every site is checked against that canonical sequence (see below),
278 and the UniProt accession is released alongside every site so that positions can
279 be verified independently.

280 3.6. Substrate and kinase identity resolution

281 Resolving protein identity from gene symbols alone is unreliable for this appli-
282 cation, because many legacy protein names are aliases of several distinct genes.
283 We therefore resolve substrate identity from the UniProt accession supplied
284 by the specificity atlases, falling back to the gene symbol only where it maps
285 unambiguously to a single gene. Kinase identity is resolved from the atlas column
286 names (which use Manning/CORAL-style short names such as P38A, CK1D and
287 MEK1 rather than HGNC symbols) against a curated kinome reference table,
288 then against unambiguous aliases, with a small explicit override list for names
289 that neither source covers; the pipeline halts if any atlas kinase name remains
290 unresolved rather than silently discarding it. Every released kinase-substrate-site
291 triplet is checked automatically against the source atlas to confirm that the site
292 belongs to the accession it is reported under. Every site is additionally checked
293 against its canonical UniProt sequence, which also yields the -7/+7 flanking
294 sequence shown for each site (verified for 95.4% of sites; unverified sites are left
295 without one rather than shown an unverified guess).

296 3.7. Confidence scoring

297 Alongside the binary criteria above, each predicted interaction carries a con-
298 tinuous confidence score. Motif percentile and functional score are each passed
299 through a monotonic, continuously differentiable logistic transform centred on
300 the corresponding threshold, and localization evidence is expressed as a saturat-
301 ing function of the number of shared HPA compartments rather than a binary
302 flag. The composite score is the unweighted mean of the three components, and
303 all three components are released as separate columns so that users can reweight
304 or threshold them independently. This is a transparent re-weighting rather than
305 a calibrated probability: a fully probabilistic treatment would require labelled
306 negative examples, which do not exist at kinome scale [26, 27].

307 3.8. Information content of the evidence layers

308 To quantify how much each evidence layer contributes and how far the layers
309 overlap, we recorded the full joint contingency table of the three indicators
310 (co-localization, curated-site status and functional-score support) over all motif-
311 passing kinase-site pairs, before any of them had been applied. From this table we
312 computed marginal Shannon entropies, pairwise mutual information, normalised
313 redundancy (mutual information divided by the smaller of the two marginal
314 entropies) and conditional mutual information, together with a leave-one-filter-
315 out analysis quantifying how many pairs each layer uniquely removes.

316 3.9. Annotation of dark kinases

317 To assess how the integrated network covers dark kinases, we used the DarkKi-
318 naseTools R package from the NIH Illuminating the Druggable Genome (IDG)
319 programme to classify kinases as “understudied (dark)” or “well-studied (light)”
320 based on publication counts, functional annotations and available reagents [14].

321 This classification was mapped onto our KSI network, enabling summary statis-
 322 tics on the proportion of curated vs. predicted substrates assigned to dark kinases.
 323 We also summarised the fraction of substrates that are novel (predicted only) for
 324 each kinase family (Figure 3(A)) and quantified the fraction of kinase–substrate
 325 pairs involving dark kinases in the curated set versus the curated-plus-specificity
 326 atlas integrated network (Figure 3(B)).

327 3.10. Implementation

328 All data integration, filtering and visualisation were performed in R using the
 329 tidyverse ecosystem for data manipulation and plotting. Kinome annotations and
 330 dark/light classifications were obtained via DarkKinaseTools and associated IDG
 331 resources [14]. Upset plots showing overlap between resources (for phosphosites
 332 and substrates) were generated using the ggupset package (Supp. Figure 1(B);
 333 Figure 1(D)). Scripts used to download source datasets, perform identifier
 334 harmonisation, apply filters and generate figures are provided in GitHub to
 335 facilitate reproducibility and reuse.

336 4. Results

337 4.1. Structure and coverage of the integrated resource

338 The final resource is organised into two primary tables:

- 339 1. **Kinase-substrate pair table.** This table lists each unique kinase–
 340 substrate pair (gene symbols), with columns indicating:
 - 341 • whether the pair is supported by curated databases, motif-based
 342 predictions, or both;
 - 343 • the number, identity and directionality of phosphosites linking the
 344 kinase and substrate;
 - 345 • source databases and PMIDs for curated evidence;
 - 346 • motif-related annotations (serine/threonine vs. tyrosine atlas);
 - 347 • contextual evidence flags (co-localization, functional score above
 348 threshold); and
 - 349 • dark/light status of the kinase.
- 350 2. **Phosphosite table.** This table lists each unique substrate site (UniProt
 351 accession, residue, position) together with its assigned kinases and annota-
 352 tions, including:
 - 353 • motif-based scores and ranks for all compatible kinases;
 - 354 • flags for high-confidence motif assignments (top 10%);
 - 355 • phosphosite functional score (continuous) and functional classification
 356 (≥ 0.5 vs. < 0.5);
 - 357 • curated vs. predicted status;
 - 358 • the -7/+7 flanking sequence, verified against the canonical sequence;
 359 and
 - 360 • basic protein metadata.

Across all sources, the integrated resource covers 426 kinases and 3,901 substrate proteins (Figure 1(B)). PhosphoSitePlus and EPSD contribute the largest numbers of substrates, while the specificity atlas brings in additional substrates not present in curated databases (Figure 1(B)). At the site level, we catalogue 13,448 unique phosphosites across PhosphoSitePlus, EPSD, iPTMnet and the specificity atlas (Supp. Figure 1(A)), with substantial but incomplete overlap between resources (Supp. Figure 1(B)).

In terms of kinase-substrate pairs, the curated KSI set contains 8,364 distinct pairs, while the high-confidence motif-based predictions add 21,916 novel pairs not present in curated databases, for a total of 30,280 distinct pairs (Figure 3(C)). Most kinases gain dozens of novel substrates: the distribution of novel substrates per kinase is right-skewed, with a median of 47 novel substrates among kinases that also have curated substrates, and some kinases gaining more than 350 (Figure 3(D)).

Alt text: Comparison of kinase coverage and substrate counts between curated databases and the integrated resource. Shows a 3.6-fold increase in KSI pairs and highlights expansion for dark kinases across different families.

4.2. Functional and family-level properties

Site-level support is most frequent for tyrosine sites and for sites drawn from iPTMnet (Figure 2(B) and Supp. Figure 1(C)). Of the 3,307 substrates with site-level support, 2,471 (75%) have at least one kinase-site pair that is also co-localized and are retained (Figure 1(C); Supp. Table 4).

Kinase and substrate coverage varies across kinase families (Figure 1(E)). TK and CMGC kinases are most numerous, whereas AGC, CAMK and “other” families contribute comparable numbers of kinases but differ in substrate counts (Figure 1(E)). When focusing on novel substrates, dark kinase families show particularly high fractions of predicted-only substrates: “other” and atypical kinases have 66.6% and 62.0% of their substrates classified as novel, respectively, while CAMK and CMGC kinases have 50.7% and 37.5% (Figure 3(A)). Incorporating the specificity atlas increases the fraction of kinase-substrate pairs involving dark kinases roughly 4.6-fold, from 3.4% in curated databases alone to 15.7% in the integrated network (Figure 3(B)), underscoring the value of motif-based predictions for illuminating poorly studied regions of the kinome.

4.3. Robustness of the filtering criteria

Because both the motif and functional-score criteria are applied as thresholds, we asked how sensitive the resource is to where those thresholds are placed. Sweeping the functional-score cut-off from 0.3 to 0.7 and the motif percentile from the 80th to the 99th (Supp. Figure 2) changes the size of the high-confidence set substantially – at the 90th motif percentile, the number of retained pairs falls from 58,319 to 13,815 across the functional-score range – but leaves the qualitative conclusions intact. The proportion of retained predicted pairs involving a dark kinase varies only between 19.1% and 20.8% across that same functional-score

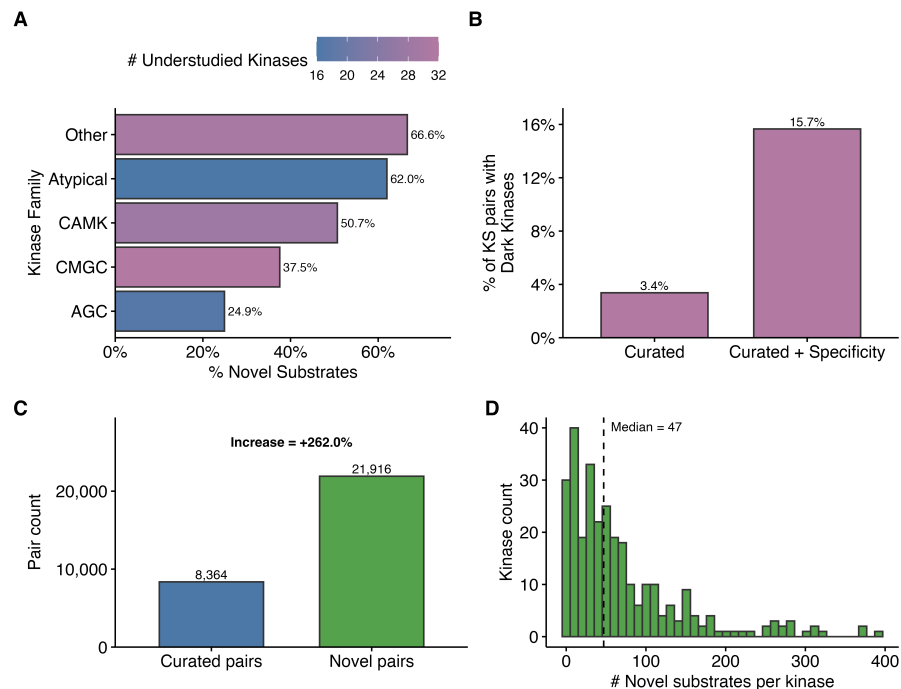


Figure 3: Expansion of the known kinase-substrate network. A) Fraction of novel substrates across major kinase families. Bars are colored by the number of dark kinases within each family, as defined by the IDG (Illuminating the Druggable Genome) classification. B) Impact of the specificity atlas on the representation of dark kinases. The bar plot shows the percentage of KS pairs involving “Dark” kinases in curated databases versus the combined “Curated + specificity atlas” dataset, demonstrating a substantial increase in the inclusion of dark signaling components. C) Global expansion of the kinase-substrate (KS) network. The bar plot compares the total number of curated pairs against novel pairs introduced by the specificity atlas, showing a 3.6-fold increase in network coverage (+262.0%). D) Histogram showing the distribution of novel substrates per kinase for kinases overlapping between curated sources and the specificity atlas. The dashed line indicates a median of 47 novel substrates per kinase.

range, and the number of kinases represented varies between 318 and 325. The sweep re-implements the filters independently; at the operating point used throughout (90th percentile, functional score 0.5) it retains 22,804 predicted pairs, against 22,797 in the released resource.

We also asked how much information each evidence layer actually contributes, and how far the layers overlap. Across all motif-passing kinase-site pairs, subcellular co-localization carries the highest marginal entropy (0.896 bits, versus 0.418 bits for curated-site status and 0.430 bits for functional-score support). Pairwise mutual information between the three layers is very small: 0.0008 bits between co-localization and curated status, 0.0007 bits between co-localization and functional-score support, and 0.0082 bits between curated status and functional-score support, giving normalised redundancies of 0.19%, 0.16% and 1.95% respectively, and conditioning on the third indicator does not change this. The layers are therefore close to orthogonal, and each is doing distinct work: in a leave-one-out analysis, co-localization uniquely removes 47,029 kinase-site pairs that satisfy every other criterion, and omitting it would inflate the high-confidence set by 182%. This filter is not trivially satisfied: the median gene has HPA annotations for two compartments and only 2.6% of annotated genes are assigned to five or more.

The specificity atlases score only phosphosites that have already been observed in phosphoproteomics experiments, and because that literature is dominated by immortalized and cancer cell lines, we expected coverage to be uneven with respect to tissue-restricted proteins. It is. Classifying substrates by HPA RNA tissue specificity, 70.4% of substrates with low tissue specificity gain at least one novel predicted interaction, falling monotonically to 57.8% for tissue-enhanced, 46.9% for group-enriched and 40.6% for tissue-enriched proteins. Conversely, the proportion of substrates for which we contribute nothing beyond existing curation rises from 29.6% to 59.4% across the same gradient. Predictions are therefore systematically sparser for specialized proteins that define differentiated cell types, and users working on tissue-restricted biology should expect the resource to be less complete for their proteins of interest.

4.4. Interactive exploration and visualization

To make these integrated data accessible to the broader community, we developed KinaseCanvas Explorer, an interactive web application that enables users to query, filter, and visualize the kinase-substrate network. The explorer provides a unified interface where users can seed the graph with a kinase or substrate of interest to generate a local interaction neighbourhood (see interface in Figure 4(A, B) and a representative network in Figure 4(C)). Key features include:

- **Kinase- or substrate-first querying:** The graph can be seeded from either a kinase or a substrate, with autocomplete on both fields. Seeding with a substrate returns all upstream kinases, curated and predicted, so

the resource can be used to ask which kinases might act on a protein of interest as readily as the reverse.

- **Site-level interrogation of a specific pair:** A pair mode restricts the view to a single kinase-substrate link and ranks the candidate sites on that substrate. This addresses the common experimental situation in which a kinase is known to phosphorylate a protein (e.g., from a band shift on a Phos-tag gel or a pan-phosphotyrosine blot) but the modified residue is unidentified.
- **Site sequence context:** Each site's -7/+7 flanking sequence is shown with the phosphoacceptor highlighted, verified against the canonical UniProt sequence for every site in the resource.
- **Quantitative evidence on every interaction:** The motif percentile, the kinase's rank for that site among all kinases, the phosphosite functional score, the number of shared subcellular compartments and the composite confidence score are all displayed in a filterable table of the interactions in view and carried into every table export. A minimum-confidence slider filters predicted edges on the composite score.
- **Explicit evidence provenance:** Curated-site status and functional-score support are reported as separate fields, so a site known to be functional from the primary literature is distinguishable from one predicted to be functional, and the view can be restricted to curated pairs or to functional sites.
- **Visual encoding:** Node shape and colour encode role (kinase, substrate, or both), and for predicted edges the line thickness and opacity scale with the composite confidence score, so relative support is visible and not merely filterable.
- **Legible neighbourhoods for promiscuous kinases:** Highly connected kinases would otherwise render as unreadable hairballs; the most connected, CDK1, has 629 substrates. A seeded neighbourhood is therefore capped at 75 interactions by default, chosen under a stated ordering (curated first, then by descending composite confidence, ties broken alphabetically), with the count shown and the cap removable in one click. The cap does not affect the 63% of kinases, and 96% of all proteins in the resource, whose neighbourhoods contain 75 or fewer interactions.
- **Click-to-inspect detail panel:** Selecting a protein shows its role, UniProt link, numbers of substrates and upstream kinases, and its highest-ranked interactions. Selecting an interaction shows its evidence, sources and PMIDs, together with a per-site table giving each site's flanking sequence, curation status, functional score, motif percentile, kinase rank and shared compartments, with curated sites listed first and the rest ranked by motif percentile.
- **Display options:** Five layout algorithms (fCoSE, CoSE, CoSE-Bilkent, Concentric and Breadthfirst) are available, together with adjustable label size and optional edge labels and arrows.
- **Export options:** Network images can be exported as PNG or PDF, and both edge-level and site-level tables can be downloaded as CSV for either

the current selection or the complete network.

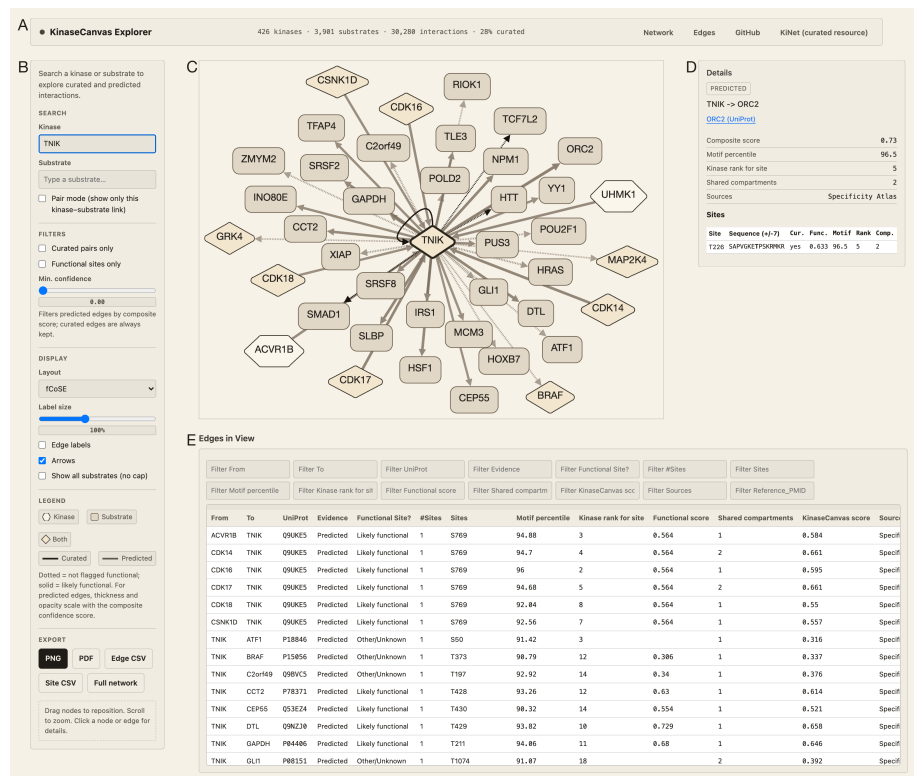


Figure 4: **The KinaseCanvas Explorer.** A) Header bar, reporting summary statistics for the released resource. B) Sidebar: kinase- and substrate-first search with autocomplete and a pair mode, filters for curated pairs, functional sites and minimum composite confidence, display options, legend and export buttons. C) Network view seeded with TNIK (TRAF2 and NCK-interacting kinase). Node shape and colour encode role (hexagon, kinase, substrate; diamond, both); edge colour distinguishes curated (dark) from predicted (light) interactions; solid lines mark sites with functional support and dotted lines sites without; and the thickness and opacity of predicted edges scale with the composite confidence score. D) Detail panel for a selected interaction (TNIK to ORC2), giving the evidence behind it and a per-site table with the flanking sequence, curation status, functional score, motif percentile, kinase rank and number of shared compartments. E) Filterable table of the interactions currently in view, which can be exported as CSV together with the site-level table.

Alt text: The KinaseCanvas Explorer interface, showing the header statistics, the search and filter sidebar, a TNIK-seeded network, the detail panel for a selected interaction with its per-site evidence, and the filterable table of interactions in view.

5. Discussion

We have developed a comprehensive resource that combines literature-curated phosphorylation data with empirically measured, kinome-wide substrate specificity profiles to chart the human kinase–substrate network. This addresses a major gap in cell-signalling knowledge: despite decades of study, substrates are still unknown or poorly defined for most human kinases [9, 10, 14]. By leveraging recent motif profiles for ~90% of the human kinome [15, 16], we systematically expand substrate lists for hundreds of kinases and then constrain these predictions with biologically motivated filters for co-localization and phosphosite functionality. The result is a collection of KSI hypotheses that is both broad—increasing the number of candidate KSIs 3.6-fold relative to curated databases alone (Figure 3(C))—and enriched for plausible, potentially regulatory interactions (Figure 1(A); Figure 2(A); Supp. Figure 1(A)).

5.1. Relationship to existing resources

Traditional resources such as PhosphoSitePlus, EPSD and iPTMnet grow as new experimental data and curated literature are added [23, 7, 8]. These databases provide high-confidence KSIs, but their coverage is limited by experimental throughput and publication bias towards a small set of kinases and pathways [14]. The recently introduced KiNet web portal builds directly on these curated datasets by aggregating interactions from PhosphoSitePlus, EPSD and iPTMnet, annotating kinases with group membership and pathway/domain context, and providing interactive network visualisations for user-selected protein sets [24].

Our work is complementary to, and explicitly builds on, KiNet. We use the KiNet-integrated KSI network as the starting curated set, but then extend it in three major ways. First, we overlay kinome-wide, empirically measured substrate specificity atlases for serine/threonine and tyrosine kinases [15, 16] to infer motif-supported kinase–phosphosite edges that are not yet observed experimentally (Figure 3(C)). Second, we integrate orthogonal contextual information: subcellular localization from HPA [22] and phosphosite functional scores from Ochoa et al. [5] to filter and prioritize predicted interactions (Figure 1(A); Figure 2(A); Supp. Figure 1(A)). Third, we systematically quantify how these additions expand coverage for dark kinases and for different kinase families (Figure 3 (A, B)). In practical terms, whereas KiNet focuses on exploring known KSIs across pathways and protein sets, our resource aims to “fill in” missing edges by providing a ranked, context-aware set of candidate interactions centred on phosphosite-level properties.

The Kinase Library web portal [15] serves as the primary interface for the Johnson et al. and Yaron-Barir et al. atlases, allowing users to score specific peptide sequences against kinase motifs. However, it is designed for sequence-level queries rather than network exploration. Similarly, tools like the KSEA App enable kinase-substrate enrichment analysis but often rely on curated databases or require users to upload custom specificity matrices, and do not provide a browsable network of high-confidence predictions.

Earlier motif-based tools such as NetPhorest, Scansite and GPS used consensus sequence motifs and position-specific scoring matrices compiled from smaller sets of peptide library experiments or literature-derived substrates [18, 17, 28]. These resources proved useful for hypothesis generation but were limited in kinase coverage and often relied on consensus motifs shared across related kinases. The specificity atlases used here are broader in scope, profiling hundreds of human kinases with oriented peptide libraries and scoring essentially the entire known human phosphoproteome [15, 16]. Intersecting these empirically derived motif scores with orthogonal features is intended to improve the signal-to-noise ratio relative to motif scores alone. This multi-layer integration is aligned with a broader trend in systems biology toward combining complementary datasets and prior knowledge to infer more accurate signalling networks [5, 16, 11].

5.2. Motif models derived from peptide libraries and from *in vitro* phosphoproteomics

The motif layer of this resource is built on positional-scanning peptide library data [15, 16], which has inherent limitations. Peptide libraries measure the intrinsic preference of the catalytic cleft against a degenerate synthetic background, which gives uniform coverage across the kinome but can have modest signal-to-noise for individual kinases and, by construction, reports only what the active site sees. An orthogonal strategy is to phosphorylate dephosphorylated proteome extracts with individual recombinant kinases and read out the products by mass spectrometry, as done by Sugiyama et al. for 385 kinases [29]. This design presents kinases with protein substrates in native sequence context, and several groups have since converted the resulting data into position-specific matrices suitable for scoring. Poll et al. derived position-dependent Shannon information matrices for 384 kinases [30], and Invergo et al. built position-specific scoring matrices and naive Bayes classifiers from the same source [26].

These two experimental designs are complementary rather than competing. Poll et al. compared their mass-spectrometry-derived “logos” directly against the peptide-library motifs used here and reported a high degree of concordance, concluding that the best predictions are likely to come from combining both data types [30]. A systematic head-to-head comparison of peptide-library and *in vitro* phosphoproteomic specificity models is an important future direction as an independent benchmarking enterprise. In the meantime, users should understand the motif layer of KinaseCanvas as one particular, uniformly measured view of intrinsic specificity rather than as a complete model of substrate recognition.

5.3. Orthogonality and information content of the evidence layers

A reasonable concern about any resource that stacks filters is whether the layers are independent or largely re-encode the same signal. The analysis in Results shows that they are close to orthogonal, and the small redundancy that does exist, between curated status and functional score, is consistent with both depending on how thoroughly a site has been observed. This is expected from how the functional score is built. It is trained on 59 features spanning

582 mass-spectrometry evidence, phosphosite regulation, structural environment and
583 evolutionary conservation [5], and subcellular localization is not among them.
584 The co-localization filter therefore contributes essentially independent evidence
585 and does substantial work of its own, rather than rubber-stamping decisions
586 already made by the other layers.

587 5.4. *Biological insights from the integrated network*

588 The integrated network reveals several notable features of kinase-substrate
589 annotation. First, the distribution of novel substrates across kinases is highly
590 non-uniform (Figure 3(D)). Many kinases with already rich substrate annotations
591 gain additional predicted substrates but do not dominate the novel interaction
592 set, partly because of our promiscuity and functional filters. In contrast, a
593 large number of kinases with few or no curated substrates acquire predicted
594 substrates, often with strong motif support, high phosphosite functional scores
595 and clear co-localization with substrates (Figure 1(A); Figure 2(A); Figure 3(C)).
596 This pattern is particularly evident among dark kinases, for which more than
597 half of substrates are novel predictions in several kinase families (Figure 3(A)).
598 The fraction of KSI pairs involving dark kinases increases from 3.4% in curated
599 databases alone to 15.7% in the integrated network (Figure 3(B)), highlighting
600 the potential of motif-informed approaches to illuminate poorly characterised
601 parts of the kinome [14].

602 Second, our residue-level analyses show that tyrosine sites are both less frequent
603 and more strongly enriched for site-level support than serine or threonine sites:
604 36.1% of tyrosine sites in our resource are either already curated or carry
605 a functional score above the threshold, compared with 12.8% of serine and
606 15.0% of threonine sites (Figure 2(B)); the corresponding functional-score-only
607 figures, among sites carrying a score, are 26.5%, 7.7% and 8.6%. We report
608 this as a property of the assembled resource rather than as a claim about the
609 phosphoproteome, because several factors plausibly contribute to it and we
610 cannot separate them here. Much of the available phosphotyrosine data was
611 generated using anti-phosphotyrosine immunoaffinity enrichment [31], a targeted
612 design that both concentrates on sites of prior biological interest and differs
613 systematically from the untargeted enrichment used for serine and threonine
614 sites. Phosphotyrosine is also the smallest and best-studied of the three classes,
615 so its coverage is comparatively closer to saturation, and a substantial share of
616 well-characterised tyrosine sites are receptor tyrosine kinase autophosphorylation
617 sites whose documented function is to recruit SH2- and PTB-domain effectors
618 [32]. Any of these would raise the apparent functional-score enrichment of
619 tyrosine sites independently of underlying regulatory importance. Published
620 analyses likewise report that tyrosine and serine/threonine phosphorylation
621 differ in their regulatory character and in the depth to which each has been
622 sampled [3, 5], and the relative evolutionary conservation of phosphothreonine
623 and phosphotyrosine sites remains an area of active disagreement [4, 5]. We
624 therefore draw no evolutionary inference from this observation.

625 The value of layering orthogonal evidence is illustrated by RANGAP1 S442,
 626 an experimentally observed mitotic phosphosite that lies in a cyclin-dependent
 627 kinase (Cdk) consensus motif but has no assigned upstream kinase in the curated
 628 databases we integrated [33]. In the serine/threonine specificity atlas [15], five
 629 kinases score at or above the 90th percentile for this site: CDK2 (rank 1, 98.8th
 630 percentile), CDK3 (rank 2, 98.1st), CDK1 (rank 3, 94.8th), MARK3 (rank 4,
 631 90.6th) and GSK3B (rank 5, 90.2nd) – kinases with overlapping motif preferences
 632 that sequence information alone cannot separate. Localization does separate
 633 them: CDK1 and CDK3 share the cytosol with RANGAP1, whereas CDK2 and
 634 GSK3B are annotated only to compartments that RANGAP1 does not occupy
 635 and MARK3 has no HPA annotation. The site also carries a high functional
 636 score (0.92). The pipeline therefore retains CDK1 and CDK3, and its preference
 637 for CDK1 over CDK2 agrees with the original report, which proposed on the
 638 basis of the timing and location of RanGAP1’s mitotic phosphorylation that
 639 Cdk1, rather than the G1/S kinase Cdk2, is the likely *in vivo* kinase, although in
 640 that study recombinant Cdk2 phosphorylated T409 far more efficiently than S442
 641 *in vitro* [33]. Because CDK1 is already curated as a RANGAP1 kinase at the
 642 neighbouring site T409 (EPSD, iPTMnet and PhosphoSitePlus), this prediction is
 643 best read as a site-level extension of an established kinase-substrate relationship,
 644 of the kind the pair mode of the Explorer is designed to support. Localization
 645 is nevertheless a static, annotation-based filter, and here it removes a kinase
 646 that is capable of the reaction: recombinant cyclin A/Cdk2 also phosphorylates
 647 RanGAP1 *in vitro* [33].

648 Similarly, our resource illuminates potential regulatory roles for dark kinases.
 649 MAPK15 (ERK8) is classified as understudied by the IDG programme [14] and
 650 has few annotated substrates despite belonging to a well-characterised, proline-
 651 directed kinase family. Our pipeline nominates RING1, the E3 ubiquitin ligase
 652 subunit of the Polycomb repressive complex 1, as a candidate substrate at S38.
 653 The prediction satisfies every criterion in the pipeline and is internally coherent
 654 with what is known about the kinase: MAPK15 is the top-ranked kinase of the
 655 entire kinome for this site (rank 1, 99.0th percentile), the site carries a high
 656 functional score (0.89), MAPK15 and RING1 share two annotated compartments
 657 (cytosol and nucleoplasm), and the local sequence (DGTEIAV-S-PRSLHSE)
 658 is a canonical serine-proline motif of the kind MAPK-family kinases require.
 659 Six of the seven kinases that score at or above the 90th percentile for this site
 660 are retained (PINK1, ranked third, shares no compartment with RING1), and
 661 MAPK15 ranks first among them, well ahead of ERK1 (MAPK3; rank 2, 93.5th
 662 percentile). S38 is an experimentally observed phosphosite that currently has
 663 no assigned upstream kinase in the curated databases we integrated, which is
 664 precisely the gap this resource is intended to fill. We present it as a prioritised
 665 hypothesis for experimental testing rather than as an established mechanism.

666 Third, the family-wise patterns in Figure 1(E) and Figure 3(A) imply that
 667 certain kinase groups such as atypical and “other” kinases may be systematically
 668 under-annotated at the substrate level. For these groups, more than half of
 669 substrates in our resource are novel, motif-supported predictions (Figure 3(A)).

670 This suggests that a substantial fraction of phosphorylation events currently
671 attributed to generic “proline-directed” or “acidic” kinases could, in fact, be
672 mediated by less-characterised kinases with similar motifs but distinct biological
673 roles. Experimental testing of these candidates using targeted kinase perturbation
674 combined with phosphoproteomics or substrate-trapping mutants could reveal
675 new pathways and cross-talk among kinase families [18, 19, 34].

676 5.5. *Limitations and caveats*

677 Despite the stringent filtering, our predictions remain hypotheses and not
678 confirmed in vivo interactions. Several limitations merit consideration. First,
679 kinases with highly similar motifs (e.g., AKT vs. SGK, or different MAPKs) will
680 often share predicted substrates [15, 32, 35]. Sequence-based motif scoring alone
681 cannot always discriminate which kinase is responsible for a given phosphosite
682 in a particular cell type or condition. Our co-localization filter removes obvious
683 mismatches (e.g., a mostly cytosolic kinase and an exclusively nuclear substrate),
684 but does not capture dynamic relocalization, scaffold-mediated proximity or
685 compartment-specific isoforms. As a consequence, some true interactions may be
686 excluded, while some retained interactions may occur only in specific contexts
687 not captured by current localization data [22]. Users should therefore treat the
688 predicted kinase list for each phosphosite as a set of candidates to be prioritised
689 rather than a definitive assignment.

690 Second, we consider each phosphosite largely in isolation, whereas substrate
691 recognition in vivo depends on determinants that lie outside the +/-7 residue
692 window a position-specific matrix can see [32]. Kinases are recruited to substrates
693 through docking interactions distinct from the catalytic cleft, including the
694 D- and DEF-motifs used by MAP kinases and the SH2- and PTB-domain
695 interactions that bring receptor tyrosine kinase substrates to high effective local
696 concentration [32]. A peptide that scores poorly against a kinase’s linear motif
697 can therefore still be an efficient substrate once bound. Recognition is also
698 frequently hierarchical: casein kinase 1 requires a phosphorylated residue at
699 n-3, GSK3 at n+4, and casein kinase 2 can use either canonical or phospho-
700 determinants, so a site’s suitability depends on prior modification events that a
701 static motif score does not represent [32]. A site that matches a kinase motif
702 may equally remain unphosphorylated if it is buried in the folded protein or if
703 an upstream priming event is absent. Conversely, kinases can act on multiple
704 sites within the same substrate, and combinatorial patterns of phosphorylation
705 may be required for functional output. Our use of a phosphosite functional
706 score partially accounts for evolutionary conservation, structural features and
707 regulatory behaviour [5], but cannot fully model three-dimensional accessibility
708 or cooperative phosphorylation events. Taken together, these mechanisms bias
709 a top-percentile motif tier against precisely those kinases for which docking,
710 rather than the catalytic cleft, dominates substrate selection – which is one
711 reason we also release a continuous confidence score and a relaxed motif tier.
712 Incorporating structural information, as done in 3D-aware prediction tools, and

713 explicit modelling of docking and priming determinants would improve specificity
714 for kinases with overlapping linear motifs [32].

715 Third, every criterion is applied as a threshold, and thresholds are arbitrary at
716 their boundaries: a site scoring 0.4999 on the functional score is excluded while
717 one scoring 0.5001 is retained, and once retained the two are treated identically
718 even though a site scoring 0.99 is far better supported. The cut-offs are not
719 arbitrary in origin (0.5 is the value recommended by the authors of the functional
720 score [5], and the 90th motif percentile follows the specificity atlases [15, 16]),
721 and we mitigate their effect in two ways: the sensitivity analysis shows which
722 conclusions depend on the operating point, and the continuous confidence score
723 lets users rank rather than filter. A fully probabilistic treatment, in which each
724 layer contributes a calibrated likelihood, would be preferable, but it requires
725 labelled negative examples that do not exist at kinome scale; existing approaches
726 rely on curated positives with sampled decoys or on in vitro assay data [26, 27].

727 Fourth, our filters are spatial but not temporal. Requiring a kinase and substrate
728 to share a subcellular compartment is a necessary condition for a genuine
729 interaction, not a sufficient one: a real kinase-substrate pair should additionally
730 show coordinated behaviour over time, with substrate phosphorylation tracking
731 kinase activity across stimuli, inhibitor treatments or genetic perturbation.
732 Compendia of perturbation phosphoproteomics now make this tractable in
733 principle [36, 35], and adding a co-regulation layer – either as an additional
734 evidence component or as a filter in its own right – is the most obvious way to
735 improve the resource.

736 Fifth, coverage is systematically uneven with respect to cell-type-specific biol-
737 ogy: novel predictions reach 70.4% of substrates with low tissue specificity but
738 only 40.6% of tissue-enriched substrates. Three mechanisms contribute to this
739 result. The specificity atlases score only phosphosites already reported in phos-
740 phoproteomics experiments, a literature dominated by immortalized and cancer
741 cell lines, so a site never observed in those systems cannot enter the resource
742 however good its motif match [37]. Tissue-restricted proteins have sparser HPA
743 annotation, so co-localization can fail for missing-data rather than biological
744 reasons [22, 38]. Lastly, functional scores depend on depth of mass-spectrometry
745 sampling, which is likewise uneven. The practical consequence is that specialized
746 proteins that give differentiated cells their identity contribute the least beyond
747 existing curation. However, by the same logic, the resource is at its most useful
748 in settings where differentiated programs are lost: cancer being the clearest case,
749 and one where large phosphoproteomic cohorts already exist to test predictions
750 against. Deeper phosphoproteomics of primary tissues and better localization
751 coverage of tissue-restricted proteins would narrow this gap.

752 Finally, we focus on the presence of potential kinase-substrate relationships
753 rather than their quantitative strength or regulatory sign (activating vs. in-
754 hibitory). Integrating quantitative phosphoproteomic time series with prior KSI
755 networks has been shown to enable inference of causal directionality and sign
756 [21]. Extending our resource with such information is a promising direction but

757 beyond the scope of the current study.

758 5.6. Future directions

759 Several extensions could further enhance the resource. First, incremental
760 updates can incorporate new phosphosites, additional motif profiling experiments
761 and improved functional annotations. As kinase specificity profiling is completed
762 for remaining kinases and refined under different conditions [15, 16], our pipeline
763 can be rerun to refresh motif-based predictions and coverage. Second, dynamic
764 data could be used to refine or reweigh predicted interactions. Condition-
765 specific phosphoproteomics, kinase inhibition or knockdown experiments, and
766 proximity-labelling assays can provide direct evidence for or against particular
767 KSIs [5, 18, 34]. Such data could be integrated into probabilistic or machine-
768 learning models that jointly consider motif score, co-localization, co-expression,
769 network proximity and experimental perturbation response to yield calibrated
770 confidence estimates, as done in other kinase-substrate prediction frameworks
771 [5, 16, 18, 17, 11]. Our current filtering scheme could thus serve as a transparent
772 baseline on top of which more sophisticated predictive models are trained. Third,
773 tighter integration with existing resources would increase usability. For example,
774 predicted KSIs could be deposited into PhosphoSitePlus, SIGNOR or related
775 databases as “computationally inferred” interactions with explicit evidence codes
776 (motif-based, co-localization-supported, functional-site-supported) [23, 8, 11].
777 This would allow researchers querying a protein of interest to view predicted
778 kinases alongside curated ones. Similarly, linking our predictions with the
779 Dark Kinase Knowledgebase and other IDG tools [14] could help experimental
780 groups prioritize substrates for poorly studied kinases, accelerating efforts to
781 “de-orphanise” dark kinases.

782 Finally, there is an opportunity to exploit the resource in disease-focused studies.
783 Many of the substrates and sites in our network overlap with proteins mutated or
784 dysregulated in cancer and other diseases [23, 5, 21, 15, 17]. Combining our KSI
785 predictions with genomic or proteomic alterations from patient cohorts could
786 reveal candidate kinase-substrate axes driving pathology, and suggest biomarkers
787 or therapeutic targets whose activity might be inferred from phosphosite status.

788 6. Conclusion

789 In summary, we present a resource that substantially broadens the current
790 landscape of human kinase-substrate relationships by integrating empirical speci-
791 ficity atlases with curated phosphorylation databases, subcellular localization
792 and phosphosite functional scores. The integrated network increases the number
793 of candidate KSIs 3.6-fold, markedly increases coverage for dark kinases, and
794 enriches for sites with features indicative of regulatory function (Figure 1(A); Fig-
795 ure 2(A); Figure 3(C); Supp. Figure 1(A)). By design, the predicted interactions
796 are hypotheses that require experimental validation, but they provide a focused
797 and context-aware starting point for dissecting signalling pathways, particularly
798 in areas of the kinome that have received little prior attention. Bridging the gap

799 between large-scale phosphoproteomic data and specific kinase assignments is
800 essential for mechanistic understanding and for translating signalling knowledge
801 into therapeutic strategies [1, 7, 5, 28, 24]; our work represents a step toward
802 that goal.

803 7. Funding

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806 X.L.P.; U24 CA211000 to X.L.P. and J.J.Y.].

807 8. Conflict of Interest

808 The authors declare no competing interests.

809 9. Acknowledgements

810 We thank the contributors to the PhosphoSitePlus, EPSD, iPTMnet, and
811 Human Protein Atlas databases for providing the primary data that underlies
812 this integrated resource.

813 10. Data Availability Statement

814 The KinaseCanvas interactive resource is available at <https://cujoisa.github.io/KinaseCanvas/>, where the integrated network can be browsed
815 and the edge-level and site-level tables downloaded in full. Each released
816 kinase-substrate-site triplet carries the individual evidence values behind its
817 classification (motif percentile, the kinase’s rank for that site, promiscuity index,
818 phosphosite functional score, number of shared subcellular compartments and
819 the composite confidence score), so the inclusion criteria can be checked case by
820 case. All code used for data integration, filtering and analysis is available at
821 <https://github.com/cujoisa/KinaseCanvas>. Supplementary Tables S1-S4 (S1
822 being the complete unfiltered prediction matrix) and the relaxed motif tier
823 are deposited on Zenodo (<https://doi.org/10.5281/zenodo.22941318>); source
824 datasets are redistributed under the terms of their original licences.
825

826 11. Supplementary Data

827 Alt text: Three-panel figure showing phosphosite counts per database, the
828 percentage of functional sites across sources, and an UpSet plot visualizing source
829 overlaps.

Unfiltered specificity atlas. Motif ranks, percentiles, and promiscuity indices for the human kinome specificity atlases. Because of its size (27.8 million rows; 3.6 GB), this table is available from Zenodo (<https://doi.org/10.5281/zenodo.22941318>) as `results/unfiltered_cantley_dataset.csv`.

Supp. Table 1

Subcellular localization mapping. Protein compartmentalization data derived from the Human Protein Atlas (HPA). Provided as `Supplementary_Table_S2.csv`.

Supp. Table 2

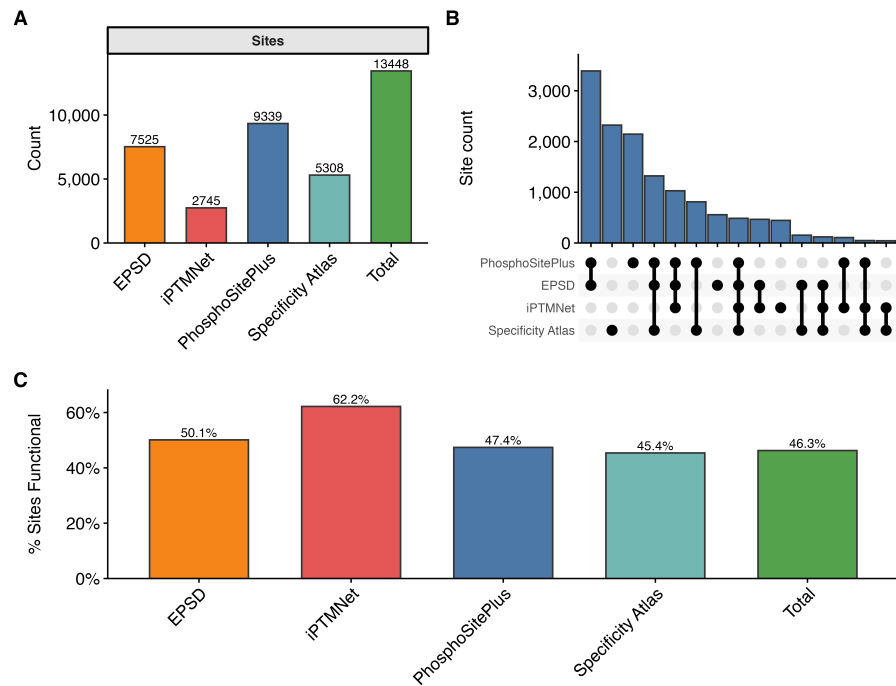
Phosphosite functional scores. Pre-computed functional scores for each reported phosphosite in the human phosphoproteome. Provided as `Supplementary_Table_S3.csv`.

Supp. Table 3

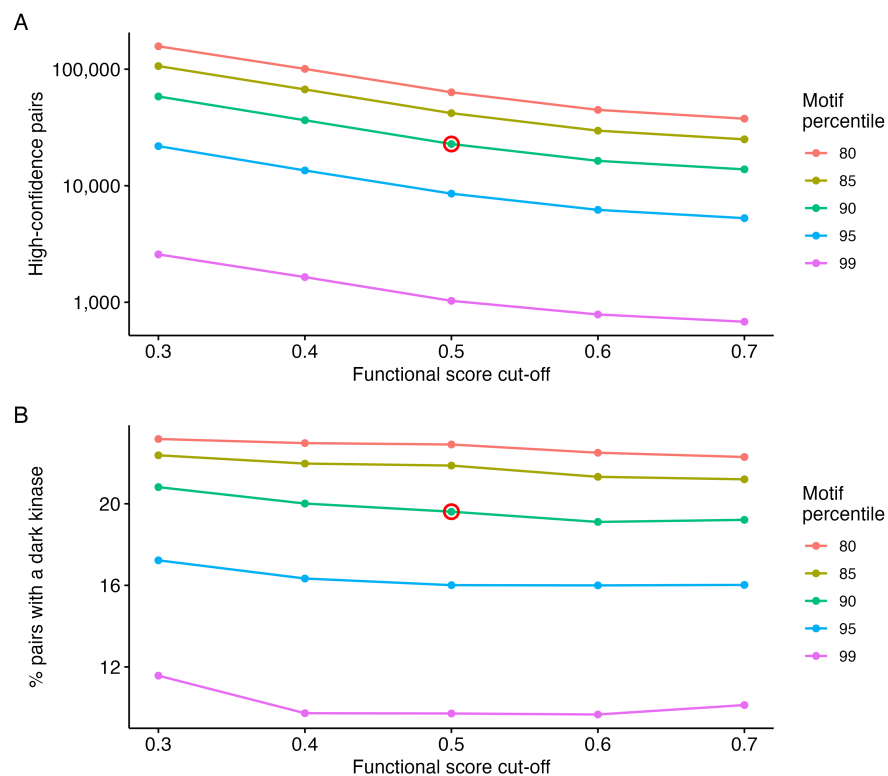
Substrate counts at each stage of the filtering summarised in Figure 1C. Each substrate is counted once per stage. A substrate enters the diagram if at least one of its sites ranks at or above the 90th percentile for a kinase with below-median promiscuity; it is “known” if it has at least one curated interaction with any kinase and “novel” otherwise. Site-level support means that at least one such site is curated or has a functional score ≥ 0.5 . A substrate is retained (“co-localized”) when at least one of its motif-passing kinase-site pairs has a supported site and a kinase that shares an HPA compartment with the substrate, so the final stage is the set of substrates contributed by the specificity atlas to the integrated resource. Substrates without site-level support are not carried into the last two stages. Provided as `Supplementary_Table_S4.csv`.

Stage	Category	Substrates
Specificity Atlas	Known substrate	2,697
Specificity Atlas	Novel substrate	6,078
Site support	Curated + functional score	800
Site support	Curated site	929
Site support	Functional score only	1,578
Site support	No site-level support	5,468
Co-localized?	Co-localized	2,471
Co-localized?	No co-localization	836
Filtered Result	Known substrate	1,602
Filtered Result	Novel substrate	869

Supp. Table 4



Supp. Figure 1: **Comprehensive landscape and functional density of integrated phosphosites.** A) Distribution of unique phosphosite counts across individual database sources (PhosphoSitePlus, EPSP, iPTMnet) and the predicted specificity atlas, alongside the non-redundant integrated total. B) UpSet plot showing the intersection of unique phosphosites across the four primary sources. While many sites are unique to specific repositories, KinaseCanvas provides a unified framework that highlights the substantial expansion provided by integrating motif-based predictions. C) Percentage of phosphosites within each source designated as 'likely functional' (Ochoa functional score ≥ 0.5), demonstrating high functional density across both curated and predicted datasets.



Supp. Figure 2: **Threshold sensitivity of the high-confidence interaction set.** A) Number of retained kinase-substrate pairs as a function of the phosphosite functional-score cut-off, at motif percentile thresholds from the 80th to the 99th. Note the logarithmic y-axis. B) Percentage of retained pairs involving a kinase classified as understudied (dark) by the IDG programme, over the same grid. The operating point used throughout the manuscript (90th motif percentile, functional score 0.5) is circled in red in both panels. The absolute size of the resource depends strongly on the thresholds chosen, whereas the dark-kinase share stays between 9.7% and 23.2% across the whole grid, well above the 3.4% of curated pairs.

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